

OLIVIA LEMAIRE

GAME PROGRAMMER / JAVA DEVELOPER

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Summary

Over 10 years of cross-disciplinary programming experience across game development and enterprise Java applications. Proficient in C++, C#, Java, and Python. Proven track record of developing scalable features, refactoring code for enhanced server performance, bolstering security through critical issue resolution, and facilitating complex data migration efforts. Experienced using modern technologies such as Unity, Unreal Engine, Vulkan, and Spring Framework. Eager to leverage technical expertise to drive innovation in software development roles.

Technical Skills

Languages: C++ · C# · Java · Python · HTML · CSS · JavaScript · SQL · XML · JSON

Technologies: Unity · Unreal Engine 4 · Git · Bitbucket · Jira · Confluence · SharePoint · Gradle · Checkmarx · Splunk

Libraries: SDL · OpenGL · Vulkan · Jakarta · Spring Framework

Experience

Scotiabank

Toronto

Java Developer

07/2024 - 09/2025

- Increased customer engagement by enabling new promotional features in 2 internal apps processing millions daily transactions.
- Improved server efficiency by refactoring code, cutting memory use, and enhancing runtime efficiency, leading to improved data handling for each transaction.
- Enhanced security posture by resolving over 100 critical errors using Checkmarx, Black Duck, SonarQube, and Splunk tools.
- Prepared new application launch by creating 2 new jobs and upgrading 5 existing jobs to enable data migration to a new database.
- Assisted in testing efforts by demonstrating the job flow of both new and existing jobs and troubleshooting issues with testing processes. Maintained detailed records of tests and their outcomes for future reference.
- Prepared for three major code deployments by merging code between different branches and manually solving conflicts.
- Supported business leaders in deciding about whether to migrate to PostgreSQL by compiling a list of all queries in our code base and analyzing the syntax to determine which would require changes if we proceeded.

Education

Humber College

Toronto

Game Programming

12/2023

Component Framework — C++ Game Engine

- Developed a 3D rendering pipeline using SDL which is compatible with OpenGL and Vulkan.
- Enabled simulations of gravity from multiple sources, rigid body collisions, kinematics, and more through a custom physics engine.
- Managed a large project making use of design principles such as entity component systems, object-oriented programming, data-driven programming, concurrency, and virtualisation.
- Allowed loading different configurations and saving data between sessions without recompiling the code using TinyXML2.

Capstone — 3D Platformer Prototype in Unity Game Engine

- Maintained steady progress by holding weekly meetings with team members to report on progress and decide on priority tasks for the next development period.
- Recorded plans and technical requirements for the project in a game design document.
- Avoided missed deadlines by assisting team members in solving problems and troubleshooting code.
- Enabled proper transition between game scenes and facilitated interactions between NPCs and the player using C# scripting.

Artificial Intelligence Project

- Integrated Kinematic Seek and Kinematic Arrive algorithms into existing physics code to allow for steering and smooth movement of actors to dynamic positions.
- Utilized decision tree logic to control NPC behaviour.
- Created a grid map system and implemented pathfinding using the A-Star algorithm.